

Course Code: BCC 103
Course Title: Business Mathematics
Marks: 100
Credit: 3 hours
(Written: 75, Internal Evaluation: 20 and Viva: 05)

Course Objectives:

The main objective of this course is to build up the mathematical knowledge of the students related to business decision making, moreover this course will provide an understanding of basic mathematical concepts and their application in the field of business. After completing this course the students will be able to take concrete decision about those problems while they face in day to day business operations.

Course Contents:

- 1. Introduction to Business Mathematics:** Importance, Scopes, Objectives and Limitations of mathematics in business studies.
- 2. Number System, Indices, Surds and Logarithm:** Origin and concept of number, Classification and properties of real & imaginary numbers, Odd number, Prime and composite numbers, Indices, Laws of indices, Uses of indices, Surds, Logarithm, Properties of Logarithm, Types of logarithm.
- 3. Functions, Equations and Inequalities:** Definitions of variable, constant, parameter and function, Domain and range, types of mathematical and business functions, Graphical representation of different types of functions, Definition of Equations, Types of Equations, Solution methods of linear and non-linear equations, Definition of Inequalities, Properties of inequalities, Solution of inequalities, Business Application.
- 4. Co-ordinate Geometry:** Abscissa and ordinate of a point, Rectangular co-ordinates, locus of a moving point and its equations, Distance between two points, Various forms of equation of a straight line, Intersection and angle between two lines, Business Application.
- 5. Set Theory:** Concept, Definition, Characteristics of Set, methods of description of set, Types of Set, Basic Set Operations, Ordered pair, Venn diagram, Cartesian Products, Business Application.
- 6. Permutations, Combinations and Progression:** Fundamental rules of counting, Permutations, Factorial notation, Permutation of 'n' Different Things, Circular Permutations, Permutations of Things not all different, Restricted Permutations, Combination, Complementary Theorems, Restricted Combinations, Combinations of things not all different, Arithmetic progression, Sum of series in Arithmetic Progression, Arithmetic Mean, Business Application, Geometric Progression, Sum of series in Geometric Progression, Geometric Mean, Business Application.

7. **Matrix Algebra:** Definition of a matrix, Types of matrices, Matrix operations, Definition of determinant, Types and properties of a determinant, Expansion of a determinant, Input-output Analysis, structure of an input - output model, open and closed model, Hawkins-Simon conditions, Determination of technological matrix and final demand, Business and Economic Applications.
8. **Differential Calculus:** Introduction, Rules of differentiation for single variable, Higher order derivatives, Concept of optimization, Optimization methods for single variable, Partial and total derivatives, Business Application.
9. **Integral Calculus:** The Relationship between Integral and Differential Calculus, Rules of Integral Calculus, Fundamental Theorem of Integral Calculus, Definite Integral, Application of Integral Calculus, Determination of consumers surplus and Producers surplus, Business Application.

Basic Texts:

1. Ann. J. Hughes, Applied Mathematics: For Business, Economics and the Social Science.
2. R. L. Childress, Mathematics for Managerial Decision, Second Edition.
3. Chiang, A.C., Fundamental Methods of Mathematical Economics", McGraw Hill Book Company, London.

Reference Books:

1. Business Mathematics, D.C. Sancheti and V. K. Kapoor, Current Edition.
2. C.A. Bush & Y. J. E. Young, Foundation of Mathematics with Application to Social & Management Science, Second Edition.
3. Dowling, E.T. "Introduction to Mathematical Economics McGraw Hill Book Company, London.
4. M. A. Taher, Business Mathematics, Current Edition.